Memory and State Control for LLM-Driven NPC Dialogue: Scoped Dual-Tier Memory and Pre-Commit Invariant Validation under a Paired, Oracle-Audited Evaluation

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***Abstract*—Large language models let non-player characters hold conversations, but they forget early facts, propose changes that break game rules, and cost more as context grows. We present OBSIDIAN FLASK, a dialogue engine that separates generation from state control: a dual-tier memory whose retrieval is scoped to the current location and character, and an event-sourced world state in which every proposed change must pass a symbolic validation barrier before commit. We compare it with six memory baselines at an equal budget, including the retrieval rule of the base paper, Generative Agents, in a memory-by-control factorial with eight ablations, on seven worlds and four models, and audit every committed change with an invariant oracle that shares no code with the validator. With a 7-billion-parameter backbone, it recalled 63.7% of facts 20 or more turns old on held-out worlds, against 0.0% for a rolling context and 54.2% for the base paper's rule, but less than flat vector memory (69.3%) and no more than its earlier version (63.3%). The barrier committed no state violation in probes or live play and accepted none of 26,273 unlawful random proposals, against 4.4% attack success under direct writes. Its cost is grounding: it rejected 20.0% of lawful requests that named entities by display name. Recovery was exact after all 300 simulated crashes.**

***Index Terms*—Large language models, non-player characters, long-term memory, event sourcing, runtime verification.**

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Source code, benchmark and run outputs: https://github.com/parthmital/RAG-Driven-NPC-Narrative-Engine.